

Global Unicorn Investment & Valuation Analysis

Business Analyst End-to-End Portfolio Project

1. Project Overview

The objective of this project was to analyze the global unicorn ecosystem to understand investment patterns, industry trends, valuation drivers, geographic dominance, and capital efficiency across unicorn companies worldwide.

The project was developed using an end-to-end analytics workflow involving Excel, PostgreSQL, Python, and Power BI to simulate an industry-level Business Analyst project.

2. Business Problem Statement

Investors, venture capital firms, startup accelerators, and policy makers need answers to questions such as:

- Which industries attract the highest investment?
- Which sectors create the highest valuation?
- Which countries produce unicorns faster?
- Which industries are most capital efficient?
- Does higher funding necessarily lead to higher valuation?
- Which startup ecosystems dominate globally?

The objective was to convert raw unicorn data into actionable business intelligence for investment and strategic decision-making.

3. Dataset Information

Dataset Name: Global Unicorn Companies Dataset

Dataset Size:

- Total Companies: 1,074
- Countries: 46
- Industries: 15
- Time Period Covered: 2007–2022

Dataset Fields:

- Company

- Valuation
 - Unicorn Year
 - Date Joined
 - Industry
 - City
 - Country
 - Continent
 - Year Founded
 - Funding
 - Select Investors
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4. Project Workflow

Phase 1: Excel Data Preparation

Objective

Prepare the raw dataset for analysis.

Activities Performed

Data Cleaning

- Checked missing values.
- Removed formatting inconsistencies.
- Standardized industry and country names.

Data Transformation

Created derived columns:

Valuation_B

Converted valuation values into numeric billions.

Example: \$180B → 180

Funding_B

Converted funding values into numeric billions.

Examples: \$8B → 8 \$572M → 0.572

Years_to_Unicorn

Calculated:

Years to Unicorn = Unicorn Year - Year Founded

Valuation Efficiency

Calculated:

Valuation Efficiency = Valuation_B / Funding_B

This metric measures value created per billion dollars invested.

Funding Category

Companies were classified into:

- Low Funding
- Medium Funding
- High Funding
- Unknown

Deliverables

- Clean dataset
- Business KPIs
- Pivot Table Analysis

Phase 2: SQL Business Analysis

Tool Used: PostgreSQL

Objective

Answer business questions using SQL.

Major SQL Questions Solved

Ecosystem Analysis

- Total number of unicorn companies.
- Total countries.
- Total industries.

Industry Analysis

- Industry with highest total funding.
- Industry with highest average funding.
- Industry with highest average valuation.
- Industry with highest capital efficiency.
- Industry producing unicorns fastest.

Geographic Analysis

- Countries with highest number of unicorns.
- Countries attracting highest funding.
- Countries producing unicorns fastest.

Investment Intelligence

- Top investors funding unicorns.
- Investors backing highest-valued companies.
- Most capital efficient companies.

Advanced SQL Concepts Used

- Aggregate Functions
- CASE WHEN
- GROUP BY
- HAVING
- ORDER BY
- Subqueries
- STRING_TO_ARRAY()
- UNNEST()

Phase 3: Python Exploratory Data Analysis

Tools Used:

- Pandas
- NumPy
- Matplotlib
- Seaborn

Objective

Validate SQL findings and perform exploratory analysis.

Activities Performed

Data Loading

Imported dataset using Pandas.

Data Validation

- Verified data types.
- Checked duplicates.
- Checked missing values.

Statistical Analysis

Generated descriptive statistics:

- Mean
- Median
- Maximum
- Minimum
- Standard Deviation

Visual Analysis

Funding Distribution

Studied funding concentration among unicorns.

Valuation Distribution

Analyzed valuation spread across companies.

Correlation Analysis

Measured relationship between funding and valuation.

Scatter Plot Analysis

Visualized funding versus valuation relationship.

Industry Analysis

- Average valuation by industry.
- Average funding by industry.
- Capital efficiency by industry.

Geographic Analysis

- Country-wise unicorn count.

- Country-wise funding trends.
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Phase 4: Power BI Dashboard Development

Objective

Convert analytical findings into executive-level business intelligence dashboards.

Dashboard Pages Developed

Executive KPI Dashboard

KPIs:

- Total Unicorns
- Total Countries
- Total Industries
- Average Valuation
- Average Years to Unicorn
- Total Funding

Geographic Analysis Dashboard

- Country-wise unicorn distribution.
- Country-wise funding distribution.
- Country-wise valuation comparison.

Industry Analysis Dashboard

- Industry funding comparison.
- Industry valuation comparison.
- Capital efficiency comparison.
- Growth speed comparison.

Investment Intelligence Dashboard

- Top valued unicorns.
- Top capital efficient unicorns.
- Funding vs valuation relationship.
- Funding category distribution.

Final Executive Dashboard

A single-page dashboard combining:

- KPIs
- Geographic Insights

- Industry Insights
 - Growth Trends
 - Capital Efficiency
 - Investment Analysis
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5. Key Business Findings

Ecosystem Findings

- Global unicorn ecosystem contains 1,074 companies.
- Unicorn companies are distributed across 46 countries and 15 industries.

Geographic Findings

- United States dominates global unicorn creation.
- China ranks second globally.
- India has emerged as the third-largest unicorn ecosystem.

Industry Findings

- Fintech received the highest total investment.
- Artificial Intelligence companies achieved the highest average valuation.
- Internet Software & Services demonstrated the highest capital efficiency.

Growth Findings

- Average time required to become a unicorn is approximately 7 years.
- China creates unicorns faster than most major startup ecosystems.

Investment Findings

- Higher funding generally leads to higher valuation.
 - However, several companies achieved very high valuations with comparatively lower funding, indicating exceptional capital efficiency.
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6. Business Recommendations

For Investors

- Prioritize Artificial Intelligence due to higher valuation potential.
- Focus on Internet Software companies due to superior capital efficiency.
- Continue monitoring Fintech due to sustained funding momentum.

For Governments

- Encourage startup ecosystems through funding programs and incubators.
- Improve access to venture capital.

For Startups

- Optimize capital allocation rather than focusing solely on fundraising.
 - Improve operational efficiency to increase valuation efficiency.
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7. Business Impact

This project demonstrates the ability to:

- Perform end-to-end business analysis.
 - Convert raw data into actionable insights.
 - Use SQL for business problem solving.
 - Apply Python for exploratory analysis.
 - Develop executive dashboards using Power BI.
 - Communicate findings to stakeholders.
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8. Technology Stack

- Microsoft Excel
 - PostgreSQL
 - Python
 - Pandas
 - NumPy
 - Matplotlib
 - Seaborn
 - Power BI
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9. Final Conclusion

The project successfully transformed raw unicorn company data into strategic business intelligence.

The analysis identified dominant startup ecosystems, high-growth industries, funding trends, and investment opportunities while demonstrating an end-to-end analytics workflow expected from a Business Analyst or Data Analyst in an industry environment.